Introductory Econometrics with Recitation: TA Session Week 13

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May 19, 2026

Today's agenda

1. Unique Function
   - `unique()`
2. Panel Data
   - Brief View
   - Model Specification
   - Model Estimation
   - Inference in Panel Data

Today's Dataset

- Today we will use the following CSV file:[1]
  - `guns.csv`: a panel dataset on U.S. states from 1977 to 1999 about the Shall-Issue laws.
-

```
gun <- read.csv("data/guns.csv")

gun <- gun %>%
  mutate(lnvio = log(vio))
```

[1]You can download the dataset here.

Extract distinct values: unique()

- unique() keeps only the distinct values in a vector.
- **Example:**

```
fruit <- c("apple", "banana", "apple", "orange", "banana")

unique(fruit)
```

- **Output:**

```
[1] "apple"  "banana" "orange"
```

- It is useful when we want to know what values appear in a variable.

A Brief View of Panel Data

Variable	Description
`stateid`	State ID (Alabama $= 1$, Alaska $= 2$, etc.)
`year`	Year (1977–1999)
`lnvio`	Log of violent crime rate
`shall`	$= 1$ if state has shall-issue law; 0 otherwise
`incarc_rate`	Incarceration rate last year (per 100,000)
`density`	Population per 1000 square miles
`avginc`	Real per capita income (thousands)
`pop`	State population (millions)
`pb1064`	Percent Black, ages 10–64
`pw1064`	Percent White, ages 10–64
`pm1029`	Percent male, ages 10–29

A Brief View of Panel Data

- Data:

stateid	year	vio	lnvio	...
1	77	414.4	6.026832	...
1	78	419.1	6.038110	...
1	79	413.3	6.024174	...
1	80	448.5	6.105909	...
1	81	470.5	6.153796	...
1	82	447.7	6.104123	...

- In panel data, we observe the same unit repeatedly over time.
- In this dataset, the unit is a state and the time period is a year.
- Together, `stateid` and `year` identify each observation.

Story: Carrying Guns and Crime?

- Some U.S. states have enacted laws allowing citizens to carry concealed weapons.
- Do these laws affect the crime rate?
    - Crime may decrease if criminals are deterred.
    - Crime may increase if more people carry guns.
- To study the effect, we need to account for state heterogeneity and time trends.
    - Some states have higher crime rates than others.
    - Crime rates may change over time.

Model Specification

- We begin with the simple model:

$$\ln(vio_{it}) = \beta_0 + \beta_1 shall_{it} + \varepsilon_{it}.$$

- Then we add control variables:

$$\ln(vio_{it}) = \beta_0 + \beta_1 shall_{it} + \beta_2 incarc\_rate_{it} + \cdots + \varepsilon_{it}.$$

- The coefficient of interest is β_1.

Model Specification

- Next, we add state fixed effects:

$$\ln(vio_{it}) = \alpha_i + \beta_1 shall_{it} + \beta_2 incarc\_rate_{it} + \cdots + \varepsilon_{it}.$$

- Finally, we add both state and year fixed effects:

$$\ln(vio_{it}) = \alpha_i + \lambda_t + \beta_1 shall_{it} + \beta_2 incarc\_rate_{it} + \cdots + \varepsilon_{it}.$$

- We can estimate fixed-effect models using `feols()` from the `fixest` package.

Model Estimation: Models (1) and (2)

- **Model (1): Simple OLS model**

```
model_1 <- lm_robust(
  lnvio ~ shall,
  data = gun
)
```

- **Model (2): OLS model with controls**

```
model_2 <- lm_robust(
  lnvio ~ shall + density + avginc + pop + pb1064 +
    pw1064 + pm1029,
  data = gun
)
```

Model Estimation: Model (3)

- **Model (3): One-way fixed effect model**
- We use `| stateid` to add state fixed effects.

```
library(fixest)

model_3 <- feols(
  lnvio ~ shall + incarc_rate + density + avginc +
    pop + pb1064 + pw1064 + pm1029 | stateid,
  data = gun
)
```

- The `|` operator separates regressors from fixed effects.

Model Estimation: Model (4)

- **Model (4): Two-way fixed effect model**
- We use `| stateid + year` to add state and year fixed effects.

```
model_4 <- feols(
  lnvio ~ shall + incarc_rate + density + avginc +
    pop + pb1064 + pw1064 + pm1029 | stateid + year,
  data = gun
)
```

Estimation Results

- **Output:**

	Simple	Control	OWFE	TWFE
(Intercept)	6.13 ***	4.20 ***		
	(0.02)	(0.66)		
shall	-0.44 ***	-0.34 ***	-0.05	-0.03
	(0.05)	(0.04)	(0.04)	(0.04)
density		0.10 ***	-0.17	-0.09
		(0.02)	(0.14)	(0.12)
avginc		0.00	-0.01	0.00
		(0.01)	(0.01)	(0.02)
pop		0.05 ***	0.01	-0.00
		(0.00)	(0.01)	(0.02)
pb1064		0.10 ***	0.10 **	0.03
		(0.02)	(0.03)	(0.05)
pw1064		0.03 **	0.04 **	0.01
		(0.01)	(0.01)	(0.02)
pm1029		-0.06 ***	-0.05 *	0.07
		(0.01)	(0.02)	(0.05)
incarc_rate			-0.00	0.00
			(0.00)	(0.00)
R^2	0.09	0.48		
Adj. R^2	0.09	0.48		
Num. obs.	1173	1173	1173	1173
RMSE	0.62	0.47		
Num. groups: stateid			51	51
R^2 (full model)			0.94	0.96
R^2 (proj model)			0.22	0.06
Adj. R^2 (full model)			0.94	0.95
Adj. R^2 (proj model)			0.21	0.05
Num. groups: year				23

*** p < 0.001; ** p < 0.01; * p < 0.05

Inference in Panel Data

- Do not forget to adjust standard errors under the panel data structure.
- Unlike cross-sectional data, errors may be correlated across time within the same unit.
- For example, we can cluster standard errors by state.
  - Errors are allowed to be correlated within each state.
  - Errors are assumed to be independent across states.
- In `fixest`, use `summary()` and specify the cluster variable through `vcov`.

```
summary(model_4, vcov = "iid") # assume homoscedasticity
summary(model_4, vcov = ~ stateid) # cluster by stateid
```

- By default, `feols` may already use clustered standard errors when fixed effects are included.

Inference in Panel Data

- **Output:**

```
OLS estimation, Dep. Var.: lnvio
Observations: 1,173
Fixed-effects: stateid: 51,  year: 23
Standard-errors: Clustered (stateid)
              Estimate Std. Error   t value Pr(>|t|)
shall        -0.027994   0.040717 -0.687518  0.49493
incarc_rate   0.000076   0.000208  0.365556  0.71624
density      -0.091555   0.123862 -0.739167  0.46326
avginc        0.000959   0.016493  0.058124  0.95388
pop          -0.004754   0.015229 -0.312188  0.75620
pb1064        0.029186   0.049541  0.589135  0.55842
pw1064        0.009250   0.023756  0.389374  0.69865
pm1029        0.073325   0.052473  1.397384  0.16847
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.135105     Adj. R2: 0.952971
                 Within R2: 0.056351
```